Priors

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Credit to:

- Tom Hobbs (Colorado State University)
- Mevin Hooten (University of Texas)
- Becky Tang (Middlebury College)
- Alison Ketz (Funga)
- Slides adopted or based on the 2024 Short course on Bayesian Modeling for Ecologists, funded by NSF (*Award DEB 2042028 to Colorado State University*)

Priors

- Priors: What you believe about parameter before you see the data. Written as probability distribution.
  - Vague/Flat
  - Informative

Informative Priors

- They are based on
  - Statistics reported in literature
  - Posterior distributions from previous studies
  - Meta-analysis
- I was trying to model range expansion by a Neo-native species due to climate change. I ended up using diffused informative prior based on previous studies.
  - Helps speed up model convergence

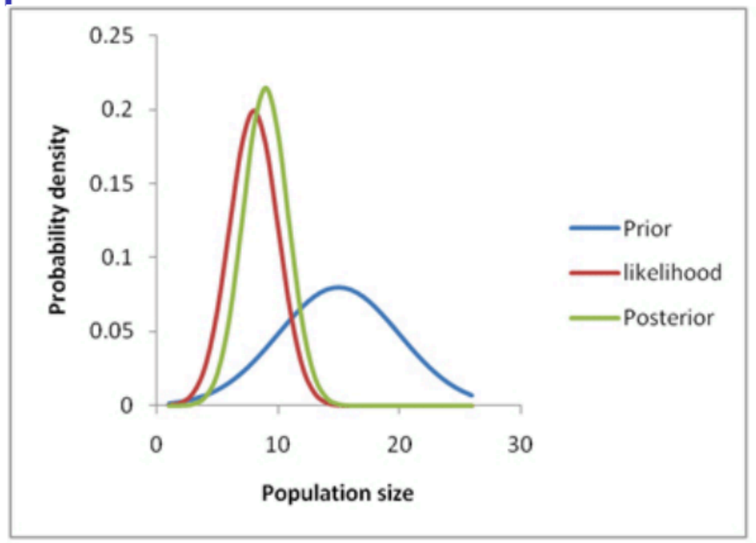
Vague or Flat Priors

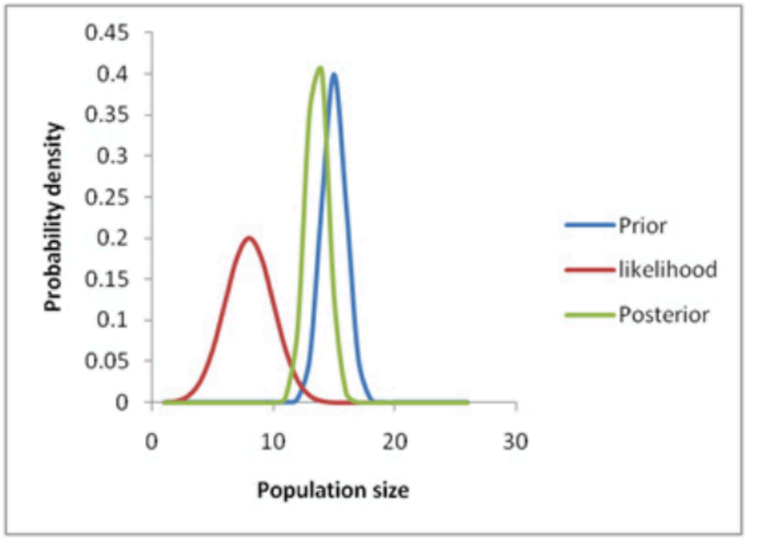
- A vague prior is a distribution with a range of uncertainity that is much larger than range of reasonable values for the parameter.

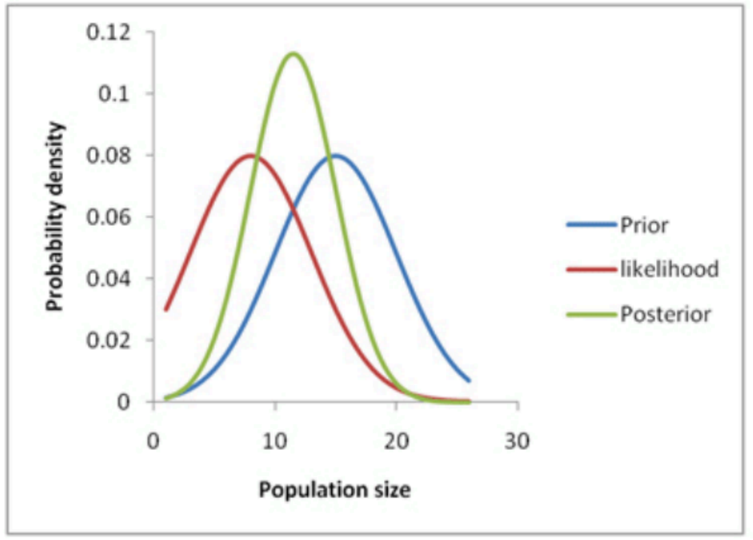
- Unlike common belief- they influence your posterior but maybe not as much as informative priors.

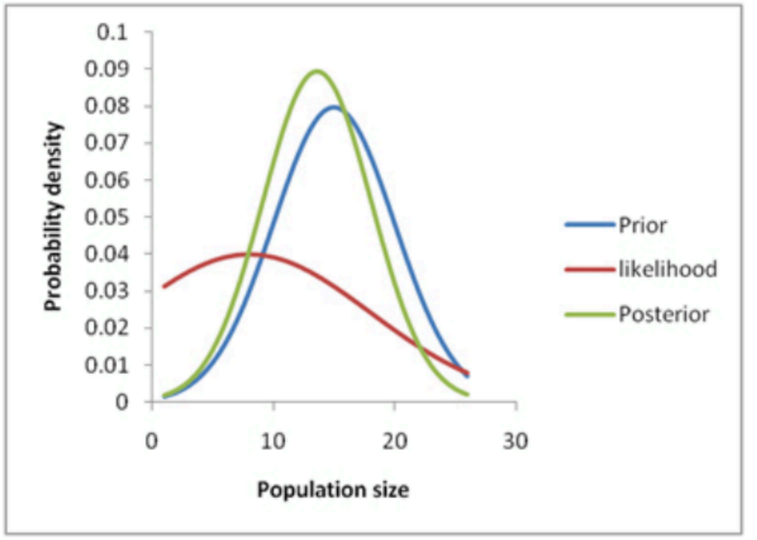
    - No prior is truly non-informative

How much does an informative prior influence the posterior?









Gündoğdu, Sedat and Akar, Mustafa. "Bayesian Update For Descriptive Statistics In Fisheries Science" Transylvanian Review

How to Choose your prior distribution

- $dx/dt = \alpha x + \beta xy$
  - x and y are the population densities of prey and predator
  - α per-capita growth rate of prey, β is the effect of predator on prey's death
    - β - cannot be positive. Needs to be negative.
      - Maybe β ~ uniform (-0.2, 0)
    - α - cannot be negative. Needs to be positive.

Choose your Vague Priors

- Some common examples:
  - For strictly non-negative quantities: Gamma (0.001, 0.001) or log-normal
  - For Variances: Uniform (0, Really large number depending on your data)
  - Regression Coeffecients: Normal (0, variance much larger than mean)
  - Survival/ Detection/Probability: Beta(1,1)

- Your priors are only vague depending on your data.
- α ~ Normal (0,1000)
  - Vague for $\alpha_0 = 2$ (prior almost flat, 0.002SD away from mean)
    - Prior density peak ~ as that of mean
  - But if your $\alpha_0 = 10{,}000$
    - 95% of its probability lies between mean ± 2SD = -2000 to 2000
    - 10,000 is 10 SD away from mean
      - Prior density peak ~ 0
        - This prior says that value of 10,000 is impossible => makes it informative prior

- Prior is steep where the data is